

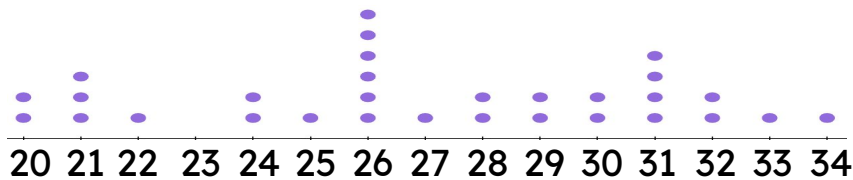


Understanding and calculating the range

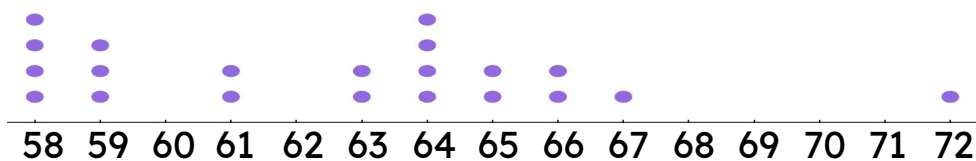
Task A: What is the range?

1) For each dot plot, work out the **range**.

a)



b)



2) For each set of data, work out the **range**.

a) -1, 0, 1, 1, 1, 2, 4, 5, 7, 7, 8, 8, 8, 10, 10, 10, 14, 17, 17, 20

b) -7, 6, 6, -3, -9, -9, -5, 6, -1, 6, 5, -6, -8, -4, -4, -3, 4, -6, 2, 6

c) 16.2, 15.3, 1.1, 19.4, 13.7, 6.7, 4.7, 12.1, 16.3, 9.4, 8.4, 18.2, 11.9, 1.6, 17.8

3)



Laura had just finished her homework on working out the **range** when her ink pen started to leak. What number has the ink spoiled in each case?

a) Largest number is [spoiled] smallest number is 14 so the **range** is 10

b) Largest number is 8, smallest number is [spoiled] so the **range** is 12

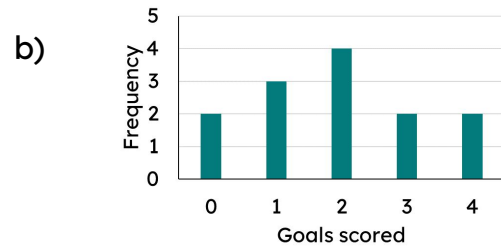
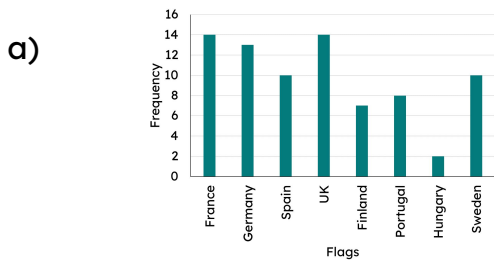
c) In order: -3, -1, -1, 0, [spoiled], 6, 7, 10 so the **range** is 13

d) Any order: 3, 5, 3, 7, 10, 12, 6, 8, [spoiled] so the **range** is 11



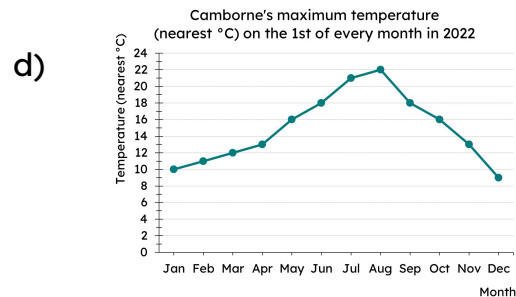
Task B: Extreme values in different representations

1) Which of these representations can you find the **range**?

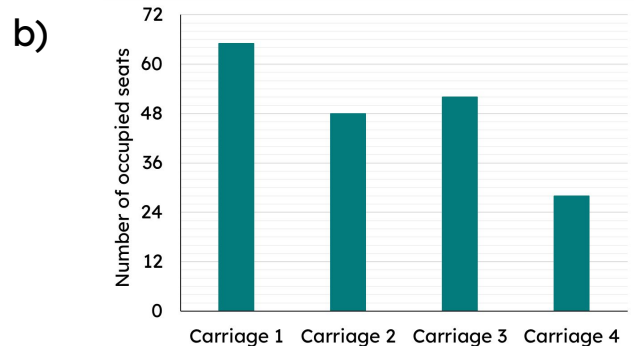
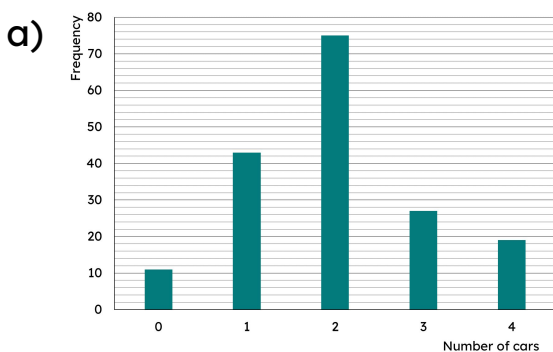


c)

Shoe size	3	4	5	6
Frequency	7	12	10	3



2) For each of the following bar charts, find the **range**.



3) Some data is represented in this frequency table about wait times at a fast food restaurant.

Wait time (mins)	4	5	6	7	8	9	10	11
Frequency	46	20	17	13	16	10	8	8

A local newspaper reported that customers' wait time **ranged** by 38 minutes. Do you agree?